

Midterm Test 2025. Row 1

1. Let $A \subseteq \mathbb{R}$ be a bounded set. Prove that **1p**

$$\forall \varepsilon > 0, \exists x \in A \text{ such that } \sup(A) - \varepsilon < x.$$

2. Study the convergence of the following series: **4p**

(a) $\sum_{n=1}^{\infty} \frac{a^n}{a+n}, a > 0.$

(c) $\sum_{n=1}^{\infty} \sqrt{n} \left(\sin \frac{1}{n} - \sin \frac{1}{n+1} \right).$

(b) $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot (2n)}.$

(d) $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\ln n}}.$

3. Compute $\int_0^{\infty} e^{-\sqrt{x}} dx.$ **1p**

4. (a) Find the Taylor series around 0 for $\sin^2 x.$ **1p**

- (b) Find the higher order derivatives $f^{(n)}(0), n \in \mathbb{N}$, for the function **0.5p**

$$f(x) = \begin{cases} \frac{1-\cos x}{x^2}, & \text{if } x \neq 0 \\ 1/2, & \text{if } x = 0. \end{cases}$$

5. Find the radius of convergence and the sum of the power series **1.5p**

$$\sum_{n=1}^{\infty} n^2 x^n.$$